

The conjectured generating function for OEIS A187588, proved

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Abstract

OEIS A187588 counts arrays of a fixed width under a local condition, and carries a conjectured generating function — and nothing else. It is true. The count is a walk count in a finite digraph on $S = 8$ vertices, so its generating function is a rational function whose numerator and denominator have degrees bounded by S ; the conjectured expression is another rational function of known degree; and two rational functions of bounded degree agree as soon as enough coefficients of their expansions do. Comparing 19 coefficients in exact arithmetic therefore settles the conjecture, rather than testing it.

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1 The conjecture

OEIS A187588 is “Number of 5-step E, S, NW and NE-moving king’s tours on an $n \times n$ board summed over all starting positions.”. It has offset 1 and begins

0, 0, 106, 578, 1484, 2794, ...

The entry states, as a conjecture contributed by a reader and never marked settled:

Empirical G.f.: $2x^3(53+130x+34x^2-15x^3)/(1-x)^3$. [Colin Barker, Jan 22 2012]

As of the “Last modified” line on the live entry (May 15 2023 14:39:30, revision 12) this is still recorded as empirical, and nothing on the entry records it as proved. The entry carries no conjectured recurrence, which is why this generating function is the whole of what is open here.

2 The count is a sum over a FIXED set of shapes

The number of steps is fixed at 5 cells, hence 4 moves, and it is the BOARD that grows. So the walks themselves do not depend on n at all. Let M be the 4 vectors the piece may move by, read from the entry’s own words, and call two walks the same SHAPE when one is a translate of the other. There are exactly 202 shapes: self-avoiding sequences of 5 distinct cells in $\mathbf{Z}^2$ with each consecutive difference in M , counted up to translation, which a finite search enumerates once and for all.

A shape occupies a bounding box with sides $(w_1, \dots, w_2)$, the range of its cells in each coordinate. Placing it on a board of side n means choosing where that box sits, and the choices in the coordinates are independent, so the shape occurs in exactly $\prod_i \max(0, n - w_i)$ positions and in no others. Distinct pairs (starting cell, walk) are exactly distinct pairs (shape, placement)

— a walk determines its shape and its translation, and conversely — so, with N_w the number of shapes of bounding box w ,

$$a(n) = \sum_w N_w \prod_{i=1}^2 \max(0, n - w_i).$$

This is exact for every n , not asymptotic, and it is not a walk count in any digraph: there is no transfer matrix here and no window.

3 The bound

Write w^* for the largest bounding-box side over all 202 shapes. Once $n \geq w^*$ every factor above is nonnegative, so the truncation does nothing and $a(n) = \sum_w N_w \prod_i (n - w_i)$, a polynomial in n of degree 2. Hence $(z - 1)^3$ annihilates a from $n = 4$ onwards, and the annihilator in force on the whole sequence is $z^5(z - 1)^3$, monic of order $S = 8$. Both numbers are read off the enumerated shapes; neither is assumed. The model was then asked for terms beyond those the entry publishes and reproduced the published ones exactly.

The family this entry belongs to is read by one model, described by its author as: “Walks of a FIXED number of steps on a board whose SIDE grows.”

The model reproduces all 35 terms the entry publishes, exactly, in integer arithmetic: it is built by reading the entry’s own English, and a misreading gives different counts, so this ties the model to the entry and not merely to the arithmetic.

4 Its generating function is rational, with bounded degree

Lemma 1. $A(x) = \sum_{n \geq 1} a(n)x^n = N_1(x)/D_1(x)$ where $D_1(x)$ has degree at most $S = 8$ and $D_1(0) = 1$, and $\deg N_1 < S + 1$.

Proof. The section above derives a MONIC linear recurrence of order S that a provably satisfies from some index onwards; write its characteristic polynomial as $z^S - \sum_{j < S} \alpha_j z^j$ and put $D_1(x) = 1 - \sum_{j < S} \alpha_j x^{S-j}$, the same polynomial written backwards, so $\deg D_1 \leq S$ and $D_1(0) = 1$. Multiplying the power series $A(x)$ by $D_1(x)$ annihilates every coefficient at which the recurrence is in force, so $N_1 = AD_1$ is a polynomial, and the indices it can be supported on are the offset together with the finitely many below the point where the recurrence takes hold, all of them less than $S + 1$. No matrix is involved: the model here is not a walk and has no adjacency matrix, and the bound comes from the recurrence the model itself supplies. $\square$

Theorem 2. *The conjectured generating function is $A(x)$.*

Proof. Write the conjecture as N_2/D_2 with $\deg N_2 = 6$ and $\deg D_2 = 3$; its expansion as a power series exists because $D_2(0) \neq 0$. Put

$$R(x) = N_1(x)D_2(x) - N_2(x)D_1(x),$$

a polynomial of degree at most 14 by Lemma 1. The difference $A - N_2/D_2$ equals $R/(D_1D_2)$, and $D_1(0)D_2(0) \neq 0$, so if the two power series agree in every coefficient up to x^{14} then R has a zero of order greater than its own degree at the origin, which forces $R \equiv 0$ and hence $A = N_2/D_2$ identically.

The coefficients of A were produced from the model by repeated matrix–vector products in exact integer arithmetic, those of N_2/D_2 by exact rational long division, and 19 of them — more than the $14 + 1$ the argument requires — agree. $\square$

5 A recurrence the entry does not state

The denominator of a rational generating function is a linear recurrence written backwards, so the theorem above yields one for free. With $D_2(x) = 1 - \sum_{i \geq 1} c_i x^i$ after normalising $D_2(0) = 1$, comparing coefficients of x^n in $A(x)D_2(x) = N_2(x)$ gives

$$a(n) = \sum_{i \geq 1} c_i a(n-i) \quad \text{for every } n > 6,$$

a constant-coefficient linear recurrence of order 3, valid past the degree of the numerator. Its coefficients are the integers $c_1, \dots, c_3$ read off D_2 , the first few being 3, -3 , 1. The entry records no recurrence, so this is not a restatement of anything already there.

6 Verification

Three checks, each independent of the algebra above.

First, the model reproduces every one of the 35 published terms, with the index shift read off the entry's offset rather than fitted.

Second, the arrays themselves were enumerated for the smallest shapes the entry counts, directly from its English and with no automaton, and the counts agree. Writing that check is where the risk actually sits: an earlier version of it dropped a clause from the entry's name in three cases and disagreed with the model, and in all three the check was wrong and the model was right.

Third, the coefficient comparison was carried out over the whole range required by the degree bound rather than on a prefix, in exact arithmetic throughout.

References

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